

# no python before code

Exactly bro, this is the key confusion. Both are imported as classes, but they have different jobs.

Think of it like this:

Car Engine Class

Car Tire Class

Both are classes.

But:

Engine → makes the car move

Tire → helps the car move

Both are classes, but their purposes are different.

In your project:

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

imports a class.

And:

```
from sklearn.naive_bayes import MultinomialNB
```

also imports a class.

So yes, BOTH are classes. 

The difference is:

TfidfVectorizer

Job:

Text

↓

Numbers

Example:

"free money now"

↓

[0.5, 0.8, 0.2]

It does NOT learn.

It does NOT predict.

It simply converts text into a format that a model can understand.

MultinomialNB

Job:

Numbers

↓

Learn patterns

↓

Predict Spam or Ham

Example:

Training Data

↓

Learns spam words

↓

Future Email

↓

SPAM

This is why it is called an ML algorithm.

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What is an algorithm?

Very simple definition:

Algorithm = a step-by-step procedure for solving a problem.

Example:

Tea-making algorithm:

1. Boil water
2. Add tea
3. Add sugar
4. Serve

Those steps form an algorithm.

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In Machine Learning:

Multinomial Naive Bayes

is an algorithm because it contains mathematical rules for:

Training

↓

Learning patterns

↓

Making predictions

It has its own logic and formulas.

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So:

TfidfVectorizer

is a class, but NOT an ML algorithm.

Why?

Because it does not learn anything.

It only prepares data.

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MultinomialNB

is a class AND an ML algorithm.

Why?

Because the class is an implementation of the Naive Bayes algorithm.

When you create:

```
model = MultinomialNB()
```

you are creating an object that uses the Naive Bayes algorithm internally.

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Interview answer:

I used the Scikit-Learn library.

For feature extraction, I used TfidfVectorizer.

For classification, I used the Multinomial Naive Bayes algorithm implemented through the MultinomialNB class.

That answer is technically correct and sounds professional. 

.feature\_extraction.text and .naive\_bayes what are these

Bro, these are modules/packages inside Scikit-Learn.

Look at this import:

```
from sklearn.feature_extraction.text import TfidfVectorizer
```

Break it down:

sklearn

↓

feature\_extraction

↓

text

↓

TfidfVectorizer

Think of it like folders on your computer:

sklearn/

|

|— feature\_extraction/

| |

| |— text/

| |

```

|   └── TfidfVectorizer
|
|   └── naive_bayes/
|       |
|       └── MultinomialNB
|
|   └── tree/
|       |
|       └── DecisionTreeClassifier
|
└── ...

```

---

What is feature\_extraction?

feature\_extraction

means:

Extract useful information from data.

In your project:

Email Text

↓

"free money now"

↓

Convert into numerical features

That's called feature extraction.

So Scikit-Learn keeps text-processing tools inside:

sklearn.feature\_extraction

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What is text?

Inside feature\_extraction there are tools for different data types.

Your data is:

Email messages

which are text.

So:

sklearn.feature\_extraction.text

contains classes that work with text.

One of them is:

TfidfVectorizer

---

What is naive\_bayes?

```
from sklearn.naive_bayes import MultinomialNB
```

Breakdown:

```
sklearn
```

↓

```
naive_bayes
```

↓

```
MultinomialNB
```

naive\_bayes is a module containing Naive Bayes algorithms.

Inside it are classes such as:

```
MultinomialNB
```

```
GaussianNB
```

```
BernoulliNB
```

These are different versions of the Naive Bayes algorithm.

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Real-world analogy

Think of a hospital:

```
Hospital
```

|

```
|— Cardiology Department
```

```
|— Neurology Department
```

```
|— Orthopedics Department
```

Similarly:

```
sklearn
```

|

```
|— feature_extraction
```

```
|— naive_bayes
```

```
|— tree
```

```
|— ensemble
```

And inside each department are specialists (classes).

Example:

```
feature_extraction
```

↓

```
TfidfVectorizer
```

```
naive_bayes
```

↓

```
MultinomialNB
```

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Notes Format

sklearn → machine learning library

feature\_extraction → module containing tools that extract useful features from data

text → submodule containing text-processing tools

TfidfVectorizer → class used to convert text into numerical features

naive\_bayes → module containing Naive Bayes classification algorithms

MultinomialNB → class implementing the Multinomial Naive Bayes algorithm

👉 Use: organize related classes and functionality inside the Scikit-Learn library.

So the hierarchy in your project is:

sklearn

↓

feature\_extraction

↓

text

↓

TfidfVectorizer

sklearn

↓

naive\_bayes

↓

MultinomialNB

They're like folders that help Python find the correct class. ✅